=== src/m3/prepare_01_start.html === <section class="cover"> Real Analysis II · Semester 5 · Module 3 <h1>Prepare: The Riemann Integral</h1> Everything you need before you open the question paper, from Plus Two calculus and Real Analysis I up to Bartle & Sherbert §7.1 (7.1.1–7.1.7), §7.2 (7.2.1–7.2.13) and §7.3 (7.3.1–7.3.18): tagged partitions, the Cauchy and Squeeze criteria, step, continuous and monotone functions, the Fundamental Theorem of Calculus, Substitution, Lebesgue's criterion, Integration by Parts and Taylor's theorem.   File 1 of 3Prepare (this file): learn the ideas and proofs File 2Model paper: all 74 past questions + new questions File 3Answer key: full formal solutions Each answer in File 3 starts with topic codes such as T15 P08 M08. If a code is new to you, read that topic here first. </section> <section class="page"> <h2>How to read this file</h2> Do, don't re-read: cover the proof, write the skeleton from memory, then check every "Why" line. Definition / Theorem · learn word for wordExact wording from Bartle & Sherbert 4e. In plain words: what the symbols actually say when you strip away the notation. Skeleton · write this before you have any ideaLet \varepsilon > 0 be given. Choose $\delta_\varepsilon = $?. Let \dot P be any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon. Then \lvert S(f;\dot P) - L\rvert \le $? $ < \varepsilon. Hence f \in \mathcal{R}[a,b] with \int_a^b f = L. ∎ KnowThe hypotheses unpacked into inequalities you can use.WantThe target statement unpacked into what you must bound. Scratch · how we find δ or n (never copied into the exam)Work backwards from \lvert S(f;\dot P) - L\rvert < \varepsilon \Leftarrow C\lVert P\rVert < \varepsilon \Leftarrow \lVert P\rVert < \varepsilon/C. Proof · what you write in the exam Forward logical step starting from \varepsilon > 0. ∎Every step names the definition, theorem or inequality that justifies it.   Hypotheses · what breaks if you drop one each condition the theorem needswhat goes wrong without it, with its counterexample tag C7 Your intuition will lie to you hereA naive mental picture that clashes with the formal definition, and how to fix it. Check yourself A quick prompt to test if you can use the idea cold. The complete short answer so you can check immediately. </section>
<section class="page"> <h2>One idea for the whole module</h2> You are handed ε > 0; you must hand back a mesh bound δ_ε > 0 Cut the interval [a,b] into pieces [x_{i-1},x_i] and pick a sample point (tag) t_i in each piece. Build a rectangle of width x_i - x_{i-1} and signed height f(t_i). The sum of these rectangle areas is the Riemann sum S(f;\dot P) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}). A function f is Riemann integrable with integral L if, whatever \varepsilon > 0 an opponent hands you, you can hand back a cut-size limit \delta_\varepsilon > 0 so that every partition whose widest piece has length \lVert P\rVert < \delta_\varepsilon—and every legal choice of tags inside those pieces—produces a sum S(f;\dot P) within \varepsilon of L.

 <table class="t game"> <tr><th>Concept</th><th>Opponent picks</th><th>You pick</th><th>Allowed to depend on</th><th>Must NOT depend on</th></tr> <tr><td>f \in \mathcal{R}[a,b] (Def 7.1.1)</td><td>\varepsilon > 0</td><td>\delta_\varepsilon > 0</td><td class="dep-on">\varepsilon, f, [a,b]</td><td class="indep">partition P, tags t_i</td></tr> <tr><td>Cauchy Criterion (7.2.1)</td><td>\varepsilon > 0</td><td>\eta_\varepsilon > 0</td><td class="dep-on">\varepsilon, f, [a,b]</td><td class="indep">\dot P, \dot Q with norm < \eta_\varepsilon</td></tr> <tr><td>Squeeze Theorem (7.2.3)</td><td>\varepsilon > 0</td><td>\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]</td><td class="dep-on">\varepsilon, f, [a,b]</td><td class="indep">individual points x</td></tr> </table> The chain that carries the module Proving integrability straight from Definition 7.1.1 needs you to guess L first. To avoid guessing L, Bartle builds one chain: Definition 7.1.1 (T03) \Rightarrow Cauchy Criterion (T10) \Rightarrow Squeeze Theorem (T12) + Step functions \in \mathcal{R} (T14) \Rightarrow Continuous \in \mathcal{R} (T15) & Monotone \in \mathcal{R} (T16). Then Mean Value Theorem (P12) connects derivatives to Riemann sums to give FTC First Form (T20), while Additivity (T17) + continuity at c gives FTC Second Form (T23), unlocking Substitution (T25), Integration by Parts (T31) and Taylor's Remainder (T32). </section> <section class="page"> <h2>The chain in one table</h2> Read down: each result is proved from the ones above it. <table class="t"> <tr><th>Step</th><th>What it says</th><th>Built from</th></tr> <tr><td>T01–T03 Definition</td><td>\forall\varepsilon>0\ \exists\delta_\varepsilon>0:\ \lVert P\rVert<\delta_\varepsilon \Rightarrow \lvert S(f;\dot P)-L\rvert<\varepsilon</td><td>partitions, tags (M02, M03)</td></tr> <tr><td>T04–T07 Basic laws</td><td>unique L; finite changes don't matter; linear and order-preserving</td><td>T03, triangle ineq. (P05)</td></tr> <tr><td>T08 Boundedness</td><td>f \in \mathcal{R}[a,b] \Rightarrow \lvert f\rvert \le M (unbounded \Rightarrow \notin \mathcal{R})</td><td>T03, reverse triangle (P05)</td></tr> <tr><td>T09 Thomae</td><td>popcorn function h \in \mathcal{R}[0,1] with \int_0^1 h = 0</td><td>T03, tag splitting (M10)</td></tr> <tr><td>T10–T11 Cauchy</td><td>\lvert S(f;\dot P)-S(f;\dot Q)\rvert < \varepsilon; Dirichlet \notin \mathcal{R}[0,1]</td><td>T03, Cauchy seq. (P06)</td></tr> <tr><td>T12 Squeeze</td><td>\alpha_\varepsilon \le f \le \omega_\varepsilon with \int(\omega_\varepsilon-\alpha_\varepsilon) < \varepsilon</td><td>T10, T07</td></tr> <tr><td>T13–T14 Step fns</td><td>\int_a^b \varphi_J = d-c; \int_a^b \sum k_j\varphi_{J_j} = \sum k_j(d_j-c_j)</td><td>T03, T06, T07</td></tr> <tr><td>T15 Continuous \Rightarrow \mathcal{R}</td><td>continuous on [a,b] \Rightarrow integrable</td><td>T12, T14, UCT (P08), Max–Min (P07)</td></tr> <tr><td>T16 Monotone \Rightarrow \mathcal{R}</td><td>monotone on [a,b] \Rightarrow integrable (staircase telescopes)</td><td>T12, T14, M05</td></tr> <tr><td>T17–T19 Additivity</td><td>\int_a^b f = \int_a^c f + \int_c^b f; restrictions; oriented \int_\alpha^\beta</td><td>T10, T08, T03</td></tr> <tr><td>T20–T21 FTC 1</td><td>F cont., F'=f off finite E, f\in\mathcal{R} \Rightarrow \int_a^b f = F(b)-F(a)</td><td>T03, MVT (P12), M05</td></tr> <tr><td>T22–T24 FTC 2</td><td>F(z)=\int_a^z f is Lipschitz; f cont. at c \Rightarrow

F'(c)=f(c)</td><td>T17, T07</td></tr> <tr><td>T25 Substitution</td><td>\int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx</td><td>T20, T23, Chain Rule (P11)</td></tr> <tr><td>T26–T30 Lebesgue</td><td>f\in\mathcal{R} \iff bounded & cont. a.e.; \varphi\circ f, \lvert f\rvert, fg \in \mathcal{R}</td><td>null sets (M07), T07, T08</td></tr> <tr><td>T31–T32 Parts & Taylor</td><td>\int_a^b fG = [FG]_a^b - \int_a^b Fg; Taylor integral remainder R_n</td><td>T20, T30, Product Rule (P11)</td></tr> </table> </section>
<section class="page"> <h2>Codes and map</h2> P01–P14Plus Two, Real Analysis I, and Modules 1–2 bridge M01–M10Maths tools built from zero T01–T32Module 3 topics (Bartle §7.1–§7.3) C7a counterexample from the library Q14question 14 of the model paper <table class="t"> <tr><th>Syllabus unit</th><th>Bartle</th><th>Topics</th></tr> <tr><td>12. Riemann Integral</td><td>7.1.1–7.1.7</td><td>T01–T09</td></tr> <tr><td>13. Riemann Integrable Functions</td><td>7.2.1–7.2.13</td><td>T10–T19</td></tr> <tr><td>14. Fundamental Theorem, Substitution, Lebesgue, Taylor</td><td>7.3.1–7.3.18</td><td>T20–T32</td></tr> </table> T01Partitions, subintervals and norm (7.1 eq. (1)) T02Tags, tagged partitions and Riemann sums (7.1 eq. (2)) T03Definition of the Riemann integral; \mathcal{R}[a,b] (7.1.1) T04Uniqueness of the integral (7.1.2) T05Integrals from the definition: k, step g, h(x)=x (7.1.4a–c) T06Changing a function at finitely many points (7.1.3, 7.1.4d) T07Linearity and monotonicity of the integral (7.1.5) T08Boundedness Theorem: f\in\mathcal{R}[a,b] \Rightarrow bounded (7.1.6) T09Thomae's function is integrable with integral 0 (7.1.7) T10Cauchy Criterion for integrability (7.2.1) T11Dirichlet's function (and relatives) not integrable (7.2.2b) T12Squeeze Theorem for integrals (7.2.3, 7.2.6b) T13Elementary step functions \varphi_J (7.2.4) T14Step functions are integrable (7.2.5, 7.2.6a) T15Continuous functions on [a,b] are integrable (7.2.7) T16Monotone functions on [a,b] are integrable (7.2.8) T17Additivity Theorem on [a,c] and [c,b] (7.2.9) T18Restriction and subdivision corollaries (7.2.10, 7.2.11) T19Oriented integrals \int_\alpha^\beta f for any \alpha,\beta,\gamma (7.2.12, 7.2.13) T20Fundamental Theorem of Calculus, First Form (7.3.1) T21Examples and limits of FTC 1: when it works and fails (7.3.2) T22Indefinite integral F(z)=\int_a^z f is Lipschitz (7.3.3, 7.3.4) T23Fundamental Theorem of Calculus, Second Form (7.3.5, 7.3.6) T24When F(z)=\int_a^z f is not an antiderivative (7.3.7) T25Substitution Theorem and change of limits (7.3.8, 7.3.9) T26Null sets and "almost everywhere"; \mathbb{Q}\cap[0,1] is null (7.3.10, 7.3.11) T27Lebesgue's Integrability Criterion and examples (7.3.12, 7.3.13)

T28Composition Theorem: \varphi\circ f \in \mathcal{R}[a,b] (7.3.14) T29Absolute value: \lvert f\rvert \in \mathcal{R}[a,b] and \lvert\int f\rvert \le \int\lvert f\rvert (7.3.15) T30Product Theorem: f,g \in \mathcal{R}[a,b] \Rightarrow fg \in \mathcal{R}[a,b] (7.3.16)
T31Integration by Parts (7.3.17)
T32Taylor's Theorem with the Integral Remainder (7.3.18) </section>
<section class="page"> <h2>The whole module in one story</h2> The story in five sentences 1. A Riemann sum S(f;\dot P) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) approximates signed area using rectangles on a partition P with sample tags t_i, and f \in \mathcal{R}[a,b] means all sums with mesh \lVert P\rVert < \delta_\varepsilon cluster within \varepsilon of one number L = \int_a^b f (T01–T04). 2. Every integrable function must be bounded (T08), may be altered at finitely many points without changing its integral (T06), and obeys linearity and order (T07), though direct \varepsilon–\delta proofs work cleanly only for simple functions like constants, step functions, x, and Thomae's function (T05, T09). 3. The Cauchy Criterion (T10) and Squeeze Theorem (T12) eliminate the need to know L in advance: by trapping f between step functions (T13–T14), we prove that every continuous function (T15) and every monotone function (T16) on [a,b] is integrable, while Dirichlet's function is not (T11), and integrals split additively across subintervals (T17–T19). 4. The two forms of the Fundamental Theorem of Calculus connect \mathcal{R}[a,b] to derivatives: FTC 1 evaluates \int_a^b f = F(b) - F(a) via the Mean Value Theorem when F'=f off a finite set and f\in\mathcal{R}[a,b] (T20–T21), while FTC 2 differentiates the Lipschitz indefinite integral F(z) = \int_a^z f at any point where f is continuous (T22–T24), yielding the Substitution Theorem (T25). 5. Lebesgue's Criterion crowns the theory—a bounded function on [a,b] is Riemann integrable if and only if its discontinuities form a null set (T26–T27)—which immediately proves the Composition, Absolute Value, and Product Theorems (T28–T30), and hence Integration by Parts (T31) and Taylor's Theorem with Integral Remainder (T32). <h3>The hypotheses to check every time</h3> Master checklist across Module 3 for f \in \mathcal{R}[a,b]: f must be bounded on [a,b] (Theorem 7.1.6)1/x on (0,1] with f(0)=0 is unbounded, so \notin \mathcal{R}[0,1] C16 for FTC 1 (Theorem 7.3.1): F continuous on [a,b], F'=f off a finite set E, and f \in \mathcal{R}[a,b]K(x)=x^2\cos(1/x^2) is differentiable everywhere, but K' is unbounded so K'\notin\mathcal{R}[0,1] C19 for FTC 2 (Theorem 7.3.5): f \in \mathcal{R}[a,b] and f is continuous at cf=\operatorname{sgn} jumps at 0, so F(x)=\lvert x\rvert-1 has a corner at 0 C20 for Substitution (Theorem 7.3.8): \varphi' is continuous on [\alpha,\beta] and f is continuous on \varphi([\alpha,\beta])\varphi(t)=\sqrt t on [0,4] has \varphi'(t)=1/(2\sqrt t) blowing up at 0 for Lebesgue's Criterion (Theorem 7.3.12): f is bounded and its discontinuity set is a null setDirichlet is bounded, but discontinuous on [0,1] (not null) C7 </section>
{{toc}}
<!-- PREREQUISITES P01-P14 --> <section class="page"> PREREQUISITES · P01–P14 <h1>What you bring from earlier courses</h1> <hr>
P01<h2>Plus Two integrals: area vs

antiderivative</h2> Two ideas you treated as the same in school; this module proves why they agree. What you already know In Plus Two you used \int_a^b f(x)\,dx in two ways: (1) the signed area between y=f(x) and the x-axis on [a,b], and (2) F(b)-F(a) where F'(x)=f(x) (such as \int x^n\,dx = \frac{x^{n+1}}{n+1} and \int \sin x\,dx = -\cos x). In Real Analysis, Definition 7.1.1 defines \int_a^b f purely from rectangle sums (view 1), and the Fundamental Theorem of Calculus (Theorem 7.3.1) is the bridge proving that F(b)-F(a) gives the exact same number.
P02<h2>Sigma (\Sigma) notation and linearity of finite sums</h2> Every Riemann sum is a finite sum, so standard algebra applies without limits. Rules for \sum_{i=1}^n For any real numbers a_i, b_i and constant c \in \mathbb{R}:

\sum_{i=1}^n c\,a_i = c\sum_{i=1}^n a_i, \qquad \sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i.
 If a_i \le b_i for each i=1,\dots,n, then \sum_{i=1}^n a_i \le \sum_{i=1}^n b_i. And summing a constant 1 over n terms gives \sum_{i=1}^n 1 = n.
P03<h2>Standard finite and geometric sums</h2> Three closed-form sums used in squeeze and null-set proofs. Formulas to know cold Integers: \displaystyle\sum_{k=1}^n k = \frac{n(n+1)}{2}, so \displaystyle\sum_{k=1}^{n-1} k = \frac{(n-1)n}{2}. Squares: \displaystyle\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}. Halving geometric series: \displaystyle\sum_{k=1}^\infty \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1/2}{1 - 1/2} = 1, so \displaystyle\sum_{k=1}^\infty \frac{\varepsilon}{2^k} = \varepsilon.
P04<h2>Supremum, infimum and completeness of ℝ</h2> Every bounded non-empty set of reals has a least upper bound \sup and greatest lower bound \inf. Real Analysis I refresher If A \subseteq \mathbb{R} is non-empty and bounded above, s = \sup A means: (i) a \le s for all a \in A, and (ii) whatever \varepsilon > 0 you are handed, there is some a \in A with a > s - \varepsilon. </section>
<section class="page"> P05<h2>Triangle inequality and the "arbitrary ε" rule</h2> The workhorse inequalities of every \varepsilon–\delta proof. Three rules used in almost every proof 1. Unpacking absolute value: \lvert a\rvert < \varepsilon \iff -\varepsilon < a < \varepsilon, and \lvert S - L\rvert < \varepsilon \iff L - \varepsilon < S < L + \varepsilon. 2. Triangle and reverse triangle inequalities: \lvert a + b\rvert \le \lvert a\rvert + \lvert b\rvert and \lvert a + b\rvert \ge \big\lvert\lvert a\rvert - \lvert b\rvert\big\rvert \ge \lvert a\rvert - \lvert b\rvert. 3. Arbitrary \varepsilon rule: If x \le y + \varepsilon for every \varepsilon > 0, then x \le y. In particular, if \lvert A - B\rvert < \varepsilon for every \varepsilon > 0, then A = B. (Why? If \lvert A - B\rvert > 0, taking \varepsilon = \tfrac12\lvert A - B\rvert > 0 gives a contradiction.)
P06<h2>Cauchy sequences in ℝ converge (Bartle 3.5.5)</h2> How we prove a limit A exists without knowing A beforehand. Theorem 3.5.5 · Cauchy Convergence Criterion A sequence (a_n) of real numbers converges to

some real number A \in \mathbb{R} if and only if for every \varepsilon > 0 there is K \in \mathbb{N} such that \lvert a_m - a_n\rvert < \varepsilon for all m, n \ge K. In Theorem 7.2.1 we build a sequence of Riemann sums a_m = S(f;\dot P_m) satisfying \lvert a_m - a_n\rvert < 1/n for all m > n; that immediately forces (a_m) to be Cauchy and therefore convergent to a real number A.
P07<h2>Continuous on [a,b]: Boundedness, Max–Min, IVT</h2> Module 1 theorems on a closed bounded interval [a,b]. Theorems 5.3.2, 5.3.4, 5.3.7 If f : [a,b] \to \mathbb{R} is continuous on the closed bounded interval [a,b], then: <ul> Boundedness (5.3.2): there is M > 0 with \lvert f(x)\rvert \le M for all x \in [a,b]. Maximum–Minimum (5.3.4): f attains its minimum and maximum on [a,b] (and on every closed subinterval I_i = [x_{i-1},x_i]): there exist u_i, v_i \in I_i with f(u_i) \le f(x) \le f(v_i) for all x \in I_i. Bolzano's IVT (5.3.7): f takes every value between \min_{[a,b]} f and \max_{[a,b]} f.
P08<h2>Uniform Continuity Theorem (Bartle 5.4.3)</h2> One \delta works across the whole closed bounded interval [a,b]. Theorem 5.4.3 · Uniform Continuity Theorem If f : [a,b] \to \mathbb{R} is continuous on the closed bounded interval [a,b], then f is uniformly continuous on [a,b]: whatever \varepsilon > 0 you are handed, there is a single \delta > 0 such that for all u, v \in [a,b] with \lvert u - v\rvert < \delta, we have \lvert f(u) - f(v)\rvert < \varepsilon. In Theorem 7.2.7 we apply this with \varepsilon/(b-a) in place of \varepsilon, so that on every subinterval of length <\delta the gap \max f - \min f is Simultaneously less than \varepsilon/(b-a). </section>
<section class="page"> P09<h2>Step functions and Lipschitz functions (Bartle 5.4.9, 5.4.4)</h2> Two special classes of functions from Module 1. Definitions 5.4.9 and 5.4.4 Step function (5.4.9): \varphi : [a,b] \to \mathbb{R} is a step function if [a,b] is the union of finitely many non-overlapping intervals J_1, \dots, J_m (open, closed, or half-open) on each of which \varphi is constant. Lipschitz function (5.4.4): F : [a,b] \to \mathbb{R} is Lipschitz if there is M > 0 such that \lvert F(z) - F(w)\rvert \le M\lvert z - w\rvert for all z, w \in [a,b]. Every Lipschitz function is uniformly continuous (and hence continuous): take \delta = \varepsilon/M.
P10<h2>Monotone functions and their jumps (Bartle 5.6.4)</h2> Increasing or decreasing functions can jump, but only countably many times. Monotone functions f : [a,b] \to \mathbb{R} is increasing if x_1 \le x_2 \Rightarrow f(x_1) \le f(x_2), and decreasing if x_1 \le x_2 \Rightarrow f(x_1) \ge f(x_2). On any subinterval [x_{k-1},x_k], an increasing function automatically has its minimum at the left endpoint x_{k-1} and its maximum at the right endpoint x_k. By Theorem 5.6.4 (cited in 7.3.13(b)), the set of discontinuities of a monotone function is at most countable.
P11<h2>Derivative rules: continuity, product, chain (Bartle §6.1)</h2> Three differentiation facts from Module 2. Theorems 6.1.2, 6.1.3(c), 6.1.6 <ul> Differentiable \Rightarrow continuous (6.1.2): If F is differentiable at c, then F is continuous at c. Product Rule (6.1.3c): (FG)' = F'G + FG'. (Reversed in Theorem 7.3.17 to give Integration by Parts.) Chain Rule (6.1.6): If H(t) = F(\varphi(t)), then H'(t) = F'(\varphi(t))\,\varphi'(t). (Used in Theorem 7.3.8 for Substitution.)

P12<h2>Mean Value Theorem (Bartle 6.2.4)</h2> The single point where Module 2 enters the proof of the Fundamental Theorem of Calculus. Theorem 6.2.4 · Mean Value Theorem If F is continuous on the closed interval [x_{i-1},x_i] and differentiable on the open interval (x_{i-1},x_i), then there exists a point u_i \in (x_{i-1},x_i) such that
F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}).
 Notice that F'(u_i)(x_i - x_{i-1}) has the exact shape of one term of a Riemann sum with tag u_i!
P13<h2>Countable sets and the Archimedean property</h2> Listing a set in a sequence r_1, r_2, r_3, \dots and choosing n > 1/\varepsilon. Countability and Archimedes Archimedean Property: Given any \varepsilon > 0, there is n \in \mathbb{N} with n > 1/\varepsilon, i.e. 1/n < \varepsilon. Countable set: A set whose elements can be listed in a sequence r_1, r_2, r_3, \dots (finite or countably infinite). In particular, the rational numbers \mathbb{Q} and \mathbb{Q}\cap[0,1] are countable.
P14<h2>Dirichlet and Thomae functions (Bartle 5.1.6(g), (h))</h2> Two benchmark functions from Module 1 that test where Riemann integration succeeds and fails. Examples 5.1.6(g) and 5.1.6(h) Dirichlet's function C7: f(x) = 1 if x \in \mathbb{Q}, f(x) = 0 if x \notin \mathbb{Q}. Discontinuous at every point of [0,1] because both rationals and irrationals are dense in \mathbb{R}. Thomae's function C8: On [0,1], h(x) = 0 if x is irrational, h(0) = 1, and h(m/n) = 1/n if m, n \in \mathbb{N} have no common integer factor except 1. Continuous at every irrational in [0,1] and discontinuous at every rational in [0,1]. </section>
<!-- MATHS TOOLS M01-M10 --> <section class="page"> MATHS TOOLS · M01–M10 <h1>The ten tools built from zero</h1> <hr>
M01<h2>Telescoping sums: why middle terms vanish</h2> When you add consecutive differences (a_i - a_{i-1}), only the first and last survive. From zero · write out four terms Suppose you have numbers a_0, a_1, a_2, \dots, a_n and you add the differences (a_i - a_{i-1}):
\sum_{i=1}^n (a_i - a_{i-1}) = (\cancel{a_1} - a_0) + (\cancel{a_2} - \cancel{a_1}) + (\cancel{a_3} - \cancel{a_2}) + \cdots + (a_n - \cancel{a_{n-1}}) = a_n - a_0.
 Three places we use this in Module 3: <ul> Lengths of subintervals: If a = x_0 < x_1 < \cdots < x_n = b, then \sum_{i=1}^n (x_i - x_{i-1}) = x_n - x_0 = b - a. The pieces of a partition always add up to the total length of [a,b]. Squares (Example 7.1.4(c)): \sum_{i=1}^n \tfrac12(x_i^2 - x_{i-1}^2) = \tfrac12(x_n^2 - x_0^2) = \tfrac12(b^2 - a^2). Cubes (Exercise 7.1.14): \sum_{i=1}^n \tfrac13(x_i^3 - x_{i-1}^3) = \tfrac13(b^3 - a^3).
M02<h2>Partitions and mesh: why \lVert P\rVert \to 0 is stronger than n \to \infty</h2> The norm \lVert P\rVert is the length of the single widest subinterval, not the number of pieces. How to find the norm of a partition Given P = (x_0, x_1, \dots, x_n) on [a,b], list the n consecutive gaps \Delta x_i = x_i - x_{i-1} and pick the

largest one: \lVert P\rVert = \max\{\Delta x_1, \dots, \Delta x_n\}. For an equal partition into n pieces of equal width, every gap is (b-a)/n, so \lVert P\rVert = (b-a)/n. Here n \to \infty makes \lVert P\rVert \to 0. For an unequal partition, having many pieces does NOT force \lVert P\rVert to be small! <figure> {{fig:F04_mesh_vs_count}} <figcaption>What to notice: Both P_a=(0,0.25,0.5,0.75,1) and P_b=(0,0.1,0.2,0.3,1) cut [0,1] into n=4 pieces, but \lVert P_a\rVert=0.25 while \lVert P_b\rVert=0.7 because of the wide right-hand piece [0.3,1]. What it does not prove: Sending n\to\infty alone never proves integrability unless the widest gap \lVert P\rVert \to 0.</figcaption> </figure>
M03<h2>Tags and endpoints: one point can tag two intervals</h2> A tag t_i can be any point in [x_{i-1},x_i], including the left or right endpoint. Why "at most 2n subintervals" appears in proofs In each subinterval I_i = [x_{i-1},x_i] we pick one tag t_i \in [x_{i-1},x_i]. What if a special point c happens to be the partition point x_k between I_k = [x_{k-1},x_k] and I_{k+1} = [x_k,x_{k+1}]? Then an opponent could legally pick t_k = c as the right-endpoint tag of I_k and t_{k+1} = c as the left-endpoint tag of I_{k+1}. So a single point c can be the tag of at most 2 subintervals, and a finite set E of n points can be the tag of at most 2n subintervals. </section>
<section class="page"> M04<h2>Bounding a sum by the largest term times total length</h2> How to turn \big\lvert\sum_{i=1}^n (f(t_i)-g(t_i))\Delta x_i\big\rvert into a single simple product. Two standard sum estimates Since subinterval lengths \Delta x_i = x_i - x_{i-1} are always positive, the triangle inequality gives
\left\lvert\sum_{i=1}^n a_i\,\Delta x_i\right\rvert \le \sum_{i=1}^n \lvert a_i\rvert\,\Delta x_i \le \left(\max_{1\le i\le n}\lvert a_i\rvert\right)\sum_{i=1}^n \Delta x_i = \left(\max_{1\le i\le n}\lvert a_i\rvert\right)(b-a).
 Case 1 (small height everywhere): If \lvert a_i\rvert < \varepsilon/(b-a) on every piece, the whole sum is < \frac{\varepsilon}{b-a}(b-a) = \varepsilon. Case 2 (one point c changed, Theorem 7.1.3): If f(x) = g(x) for all x \ne c, then f(t_i) - g(t_i) = 0 except on the (at most 2) subintervals whose tag is c. Each has width \Delta x_i \le \lVert P\rVert and height \lvert f(c) - g(c)\rvert \le \lvert f(c)\rvert + \lvert g(c)\rvert, so
\lvert S(f;\dot P) - S(g;\dot P)\rvert \le 2\big(\lvert f(c)\rvert + \lvert g(c)\rvert\big)\lVert P\rVert.

M05<h2>Telescoping with monotonicity and with FTC</h2> Two places where telescoping turns n rectangles into f(b)-f(a) or F(b)-F(a). Staircase cancellation and MVT cancellation 1. Monotone staircase (Theorem 7.2.8): On n equal pieces of width (b-a)/n, an increasing function f has \max_{I_k} f = f(x_k) and \min_{I_k} f = f(x_{k-1}). The difference between upper and lower step integrals is
\frac{b-a}{n}\sum_{k=1}^n \big(f(x_k) - f(x_{k-1})\big) = \frac{b-a}{n}\big(f(b) - f(a)\big).
 2. FTC First Form (Theorem 7.3.1): Applying the MVT to F on each [x_{i-1},x_i] gives F(x_i) - F(x_{i-1}) = f(u_i)\Delta x_i. Summing over i=1,\dots,n telescopes the left side to F(b) - F(a) and turns the right side into S(f;\dot P_u).
M06<h2>Elementary step functions \varphi_J as building blocks</h2> Write any step function as \sum_{j=1}^m k_j\varphi_{J_j} and read off its integral immediately. <div class="box

maths">Indicator recipe For any subinterval J \subseteq [a,b] with endpoints c < d, the elementary step function \varphi_J(x) equals 1 on J and 0 outside J. Its integral is simply the length of J: \int_a^b \varphi_J = d - c (Lemma 7.2.4). If \varphi equals k_1 on J_1, k_2 on J_2, \dots, k_m on J_m (where J_1,\dots,J_m partition [a,b]), then \varphi(x) = \sum_{j=1}^m k_j\varphi_{J_j}(x), so \int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j): sum of (height \times width). </section>
<section class="page"> M07<h2>The \varepsilon/2^k covering trick for countable sets</h2> How an infinite sequence of points can be covered by intervals of total length \le \varepsilon. Why countable sets have zero total length Suppose Z = \{r_1, r_2, r_3, \dots\} is countable. If we gave every point r_k an interval of the same length \delta, the total length \delta + \delta + \cdots would be \infty. Instead, we shrink the interval width by half at each step! Put r_k inside the open interval J_k = \left(r_k - \dfrac{\varepsilon}{2^{k+1}},\, r_k + \dfrac{\varepsilon}{2^{k+1}}\right). The length of J_k is 2\cdot\dfrac{\varepsilon}{2^{k+1}} = \dfrac{\varepsilon}{2^k}. By P03, the total length of all infinitely many intervals is \sum_{k=1}^\infty \dfrac{\varepsilon}{2^k} = \varepsilon.
M08<h2>The \varepsilon/2 (or \varepsilon/3) argument and \delta = \min\{\delta_1,\delta_2\}</h2> How to combine two or three separate error estimates at the same time. Combining tolerances When a proof splits an error into two parts, \lvert A - B\rvert \le \text{Error}_1 + \text{Error}_2, ask for each part to be < \varepsilon/2 (or < \varepsilon/3 if there are three parts, as in the Additivity Theorem 7.2.9). If \text{Error}_1 < \varepsilon/2 needs \lVert P\rVert < \delta_1 and \text{Error}_2 < \varepsilon/2 needs \lVert P\rVert < \delta_2, choose
\delta = \min\{\delta_1, \delta_2\} > 0.
 Then any partition with \lVert P\rVert < \delta satisfies both \lVert P\rVert < \delta_1 and \lVert P\rVert < \delta_2 simultaneously, so \text{Error}_1 + \text{Error}_2 < \varepsilon/2 + \varepsilon/2 = \varepsilon.
M09<h2>Choosing \delta_\varepsilon explicitly from C\delta < \varepsilon</h2> Do the algebra in scratch work first so your proof starts with a clean choice of \delta_\varepsilon. Backwards in scratch, forwards in proof If your scratch work shows that \lVert P\rVert < \delta implies \lvert S(f;\dot P) - L\rvert \le C\delta (for instance \lvert S(g;\dot P) - 8\rvert \le 5\delta in Example 7.1.4(b)), then to force C\delta < \varepsilon you simply pick any \delta_\varepsilon < \varepsilon/C (such as \delta_\varepsilon = \varepsilon/(2C), e.g. \varepsilon/10). If your estimate isalready strict (\lvert S - L\rvert < \delta), then \delta_\varepsilon = \varepsilon works directly.
M10<h2>Splitting a Riemann sum by where the tags land</h2> How Bartle handles jumps and popcorn functions (Examples 7.1.4(b),(d), 7.1.7; Exercises 7.1.4–5). Two ways to split \dot P = \dot P_1 \cup \dot P_2 1. Tag in a subinterval [c_1,c_2] (Exercises 7.1.4–5): Suppose \lVert P\rVert < \delta and \dot P_1 consists of the subintervals I_i whose tag t_i lands in [c_1,c_2]. Because each I_i has width <\delta, no such interval can stick out past c_1-\delta or c_2+\delta, and every point of [c_1+\delta,c_2-\delta] must lie in one of them. Thus the union U_1 of intervals in \dot P_1 is trapped between:
[c_1+\delta,\, c_2-\delta] \subseteq U_1 \subseteq [c_1-\delta,\, c_2+\delta],
 so the total width of \dot P_1 lies between (c_2-c_1)-2\delta and (c_2-c_1)+2\delta (or

between 1-\delta and 1+\delta when c_1=a=0 is the endpoint of [0,3]). 2. Tag in a finite "bad" set E = \{x : f(x) \ge \varepsilon/2\} (Example 7.1.7): Put \dot P_1 = subintervals with tag t_i \in E, and \dot P_2 = subintervals with tag t_i \notin E. If E has n points, \dot P_1 has at most 2n subintervals (by M03) of total width \le 2n\lVert P\rVert, while on \dot P_2 the height is <\varepsilon/2 and total width is \le b-a. </section>
=== src/m3/prepare_02_sec71.html === <!-- UNIT 1: BARTLE §7.1 (T01-T09) --> <section class="page"> UNIT 1 · BARTLE §7.1 <h1>The Riemann Integral</h1> <hr>
<!-- T01 (K01) --> T01<h2>Partitions, subintervals and the norm</h2> Cutting [a,b] into pieces and measuring the widest piece (PYQ MTS-26 Q4, MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16). <figure> {{fig:F01_partition_norm}} <figcaption>What to notice: For P=(0,1,1.5,2,3.4,4) on [0,4], the five subinterval widths are 1, 0.5, 0.5, 1.4, 0.6; the norm \lVert P\rVert = 1.4 is the single largest gap I_4=[2,3.4]. What it does not prove: Knowing \lVert P\rVert does not tell you how many subintervals P has or where the other cut points sit.</figcaption> </figure> Definition · Partition and Norm (Bartle §7.1, p. 199–200, eq. (1))If I := [a,b] is a closed bounded interval in \mathbb{R}, then a partition of I is a finite, ordered set P := (x_0, x_1, \dots, x_{n-1}, x_n) of points in I such that
a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.
The points of P are used to divide I = [a,b] into non-overlapping subintervals
I_1 := [x_0, x_1],\quad I_2 := [x_1, x_2],\quad \dots,\quad I_n := [x_{n-1}, x_n].
We define the norm (or mesh) of P to be the number
\lVert P\rVert := \max\{x_1 - x_0,\, x_2 - x_1,\, \dots,\, x_n - x_{n-1}\}.
In plain words: a partition is a strictly increasing list of cut points starting at a and ending at b; its norm \lVert P\rVert is the length of the widest piece. Skeleton · find the norm of a given partitionList the subinterval lengths \Delta x_i = x_i - x_{i-1} for i=1,\dots,n: ?. Therefore \lVert P\rVert = \max\{?\} = $?. Worked example · Bartle Exercise 7.1.1 & Past Papers 1. On $I = [0, 4]$ (Bartle Ex 7.1.1; PYQ MTS-26 Q4, MAT-22 Q3): <ul> $P_1 = (0, 1, 2, 4)$: gaps are $1 - 0 = 1$, $2 - 1 = 1$, $4 - 2 = 2$, so $\lVert P_1\rVert = \max\{1,1,2\} = 2$. $P_2 = (0, 2, 3, 4)$: gaps are $2, 1, 1$, so $\lVert P_2\rVert = \max\{2,1,1\} = 2$. $P_3 = (0, 1, 1.5, 2, 3.4, 4)$: gaps are $1, 0.5, 0.5, 1.4, 0.6$, so $\lVert P_3\rVert = 1.4$. $P_4 = (0, 0.5, 2.5, 3.5, 4)$: gaps are $0.5, 2, 1, 0.5$, so $\lVert P_4\rVert = 2$. 2. Non-example: $(0, 2, 1, 4)$ is not a partition of $[0, 4]$ (not strictly increasing: $2 \not< 1$), and $(0, 1, 2, 3)$ is not a partition of $[0, 4]$ because $x_3 = 3 \neq 4 = b$. Check yourself · PYQ MAT-22 MCQ 16 On $I = [2, 9]$, which partition has the largest norm: $P_1 = (2, 4, 6, 8, 9)$, $P_2 = (2, 6, 9)$, or $P_3 = (2, 5, 7, 9)$? \lVert P_1\rVert = \max{2,2,2,1} = 2$; \lVert P_2\rVert = \max\{4,3\} = 4; \lVert P_3\rVert = \max\{3,2,2\} = 3. So P_2 has the largest norm (4). In the model paper: Q1 (PYQ MTS-26 Q4, MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16). </section>
<!-- T02 (K02) --> <section class="page"> <span

class="tid">T02<h2>Tags, tagged partitions and Riemann sums</h2> Picking a sample point t_i in each subinterval and adding the rectangle areas (PYQ MTS-22 Q4). <figure> {{fig:F02_tagged_riemann_sum}} <figcaption>What to notice: For f(x)=x^2 on [0,1] with P=(0,0.2,0.5,0.7,1) and tags t=(0.1,0.5,0.6,0.9), each rectangle's top touches the curve at (t_i,f(t_i)), giving S(f;\dot P)=0.002+0.075+0.072+0.243=0.392. What it does not prove: A single Riemann sum 0.392 is only an approximation, not the exact integral 1/3.</figcaption> </figure> Definition · Tagged Partition and Riemann Sum (Bartle §7.1, p. 200, eq. (2))If a point t_i has been selected from each subinterval I_i = [x_{i-1}, x_i], for i = 1, 2, \dots, n, then the points are called tags of the subintervals I_i. A set of ordered pairs \dot P := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n
of subintervals and corresponding tags is called a tagged partition of I; the dot over the P indicates that a tag has been chosen for each subinterval. The norm of a tagged partition is defined to be the same as for the corresponding ordinary partition and does not depend on the choice of tags. If f : [a,b] \to \mathbb{R} is a function defined on [a,b], then the Riemann sum of f corresponding to the tagged partition \dot P is the number S(f;\dot P) := \sum_{i=1}^n f(t_i)(x_i - x_{i-1}).
In plain words: in each piece [x_{i-1},x_i], evaluate f at its chosen tag t_i, multiply by the piece's width (x_i - x_{i-1}), and add up all n products. Skeleton · compute a Riemann sum S(f;\dot P)List the subintervals I_i = [x_{i-1},x_i], widths \Delta x_i = x_i - x_{i-1}, and tags t_i \in I_i. Compute f(t_i) for each i. Then $S(f;\dot P) = \sum_{i=1}^n f(t_i)\Delta x_i = $?. Worked example · Bartle Exercise 7.1.2 Let f(x) = x^2 on [0,4]. We compute S(f;\dot P) for P_1 = (0,1,2,4) (widths 1, 1, 2) and P_2 = (0,2,3,4) (widths 2, 1, 1): <ul> (a) P_1 with left endpoints t = (0,1,2): S = f(0)\cdot 1 + f(1)\cdot 1 + f(2)\cdot 2 = 0(1) + 1(1) + 4(2) = 9. (b) P_1 with right endpoints t = (1,2,4): S = f(1)\cdot 1 + f(2)\cdot 1 + f(4)\cdot 2 = 1(1) + 4(1) + 16(2) = 37. (c) P_2 with left endpoints t = (0,2,3): S = 0(2) + 4(1) + 9(1) = 13. (d) P_2 with right endpoints t = (2,3,4): S = 4(2) + 9(1) + 16(1) = 33. Non-example: Taking t_1 = 1.5 for I_1 = [0,1] is not a tagged partition because the tag t_1 must belong to its own subinterval [0,1]. Your intuition will lie to you here Trap 1: Writing f(x_i) in the definition of a Riemann sum. The definition requires f(t_i) for an arbitrary tag t_i \in [x_{i-1},x_i], not just the right endpoint x_i. Trap 2: Thinking S(f;\dot P) is always positive. Where f(t_i) < 0, the term f(t_i)(x_i - x_{i-1}) is negative (signed area). In the model paper: Q2 (PYQ MTS-22 Q4). </section>
<!-- T03 (K03) --> <section class="page"> T03<h2>Definition of the Riemann integral (\mathcal{R}[a,b])</h2> The \varepsilon–\delta_\varepsilon definition that governs the entire module (PYQ MAT-21 MCQ 9). <figure> {{fig:F03_four_tag_rules}} <figcaption>What to notice: On the same partition P=(0,0.25,0.5,0.75,1) of [0,1] for f(x)=x^2, left tags give 0.21875, right tags give 0.46875, midpoints give 0.328125, and random tags (0.2,0.3,0.7,0.8) give 0.315—all clustering near L=1/3. What it does not prove: Checking four tag choices on one partition does not prove f\in\mathcal{R}[0,1]; every tag choice on every partition with \lVert P\rVert<\delta_\varepsilon must lie in (L-\varepsilon,L+\varepsilon).</figcaption> </figure> Definition 7.1.1 · Riemann Integrable (Bartle p. 201) · say it word for wordA function f : [a,b] \to \mathbb{R} is said to be Riemann

integrable on [a,b] if there exists a number L \in \mathbb{R} such that for every \varepsilon > 0 there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, then
\lvert S(f;\dot P) - L\rvert < \varepsilon.
The set of all Riemann integrable functions on [a,b] is denoted by \mathcal{R}[a,b]. We write L = \int_a^b f or \int_a^b f(x)\,dx. In plain words: there is one target number L such that, whatever error tolerance \varepsilon > 0 you are handed, you can find a mesh bound \delta_\varepsilon > 0 forcing every Riemann sum with widest piece <\delta_\varepsilon (for any choice of cut points and any choice of tags) to land within \varepsilon of L. </div> <div class="box skel">Skeleton · prove f \in \mathcal{R}[a,b] with \int_a^b f = L directly from Definition 7.1.1Let \varepsilon > 0 be given. Choose \delta_\varepsilon = $? $ > 0. Let \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n be any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon. Then \lvert S(f;\dot P) - L\rvert \le $? $ < \varepsilon. Therefore f \in \mathcal{R}[a,b] and \int_a^b f = L. ∎</div> <div class="box idea">Remark 7.1.1 & Exercise 7.1.3 · why this is not an ordinary limit <p>Why \lim_{\lVert P\rVert \to 0} S(f;\dot P) is not an ordinary function limit: For a single number \delta = \lVert P\rVert, there are infinitely many partitions P having that exact norm and infinitely many tag choices t_i for each P. Thus S(f;\dot P) is not a single-valued function of \lVert P\rVert.</p> <p>Strict < versus \le (Bartle Ex 7.1.3): Replacing \lVert P\rVert < \delta_\varepsilon and \lvert S(f;\dot P) - L\rvert < \varepsilon by \le gives an equivalent definition (given \varepsilon, apply the \le-version with \varepsilon/2 and take \delta'_\varepsilon = \tfrac12\delta_{\varepsilon/2}).</p> <p>Worked example & Non-example: f(x)=5 on [0,2] is in \mathcal{R}[0,2] with L=10 since S(f;\dot P)=10 for every \dot P (any \delta_\varepsilon>0 works). By contrast, Dirichlet's function C7 on [0,1] is not in \mathcal{R}[0,1] because on every partition, rational tags give S=1 and irrational tags give S=0, so no single number L can be within \varepsilon = 1/2 of both.</p> </div> <p class="small">In the model paper: Q3 (PYQ MAT-21 MCQ 9).</p> </section>
<!-- T04 (K04) --> <section class="page"> <div class="topic-head">T04<h2>Uniqueness of the integral</h2></div> <p class="oneline">A function in \mathcal{R}[a,b] cannot have two different integrals (Bartle Theorem 7.1.2).</p> <div class="box def">Theorem 7.1.2 · Uniqueness of the Integral (Bartle p. 201)If f \in \mathcal{R}[a,b], then the value of the integral is uniquely determined. In plain words: if L' and L'' both satisfy Definition 7.1.1 for the same f on [a,b], then L' = L''. </div> <div class="box hyp">Hypotheses · Theorem 7.1.2 f \in \mathcal{R}[a,b] (so \delta_\varepsilon works for all tagged partitions of norm <\delta_\varepsilon)if you only require convergence along one chosen sequence of tagged partitions (\dot P_n), different tag sequences can converge to different limits: for Dirichlet C7, rational tags give 1 while irrational tags give 0 </div> <div class="kw">KnowL' and L'' both satisfy Definition 7.1.1 for f on [a,b].WantL' = L'', by showing \lvert L' - L''\rvert < \varepsilon for every \varepsilon > 0.</div> <div class="box scratch">Scratch · how we link L' and L''Add and subtract the same Riemann sum S(f;\dot P): \lvert L' - L''\rvert \le \lvert L' - S(f;\dot P)\rvert + \lvert S(f;\dot P) - L''\rvert. Make each piece < \varepsilon/2 by taking \delta_\varepsilon = \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\}.</div> <div class="box proof">Proof of Theorem 7.1.2 (Bartle p. 201) <div class="steps"> <div class="step">Assume that L' and L'' both satisfy Definition 7.1.1 and let \varepsilon > 0.<span

class="why">To prove L' = L'', we will show \lvert L' - L''\rvert < \varepsilon for arbitrary \varepsilon > 0 (P05). There exists \delta'_{\varepsilon/2} > 0 such that if \dot P_1 is any tagged partition with \lVert P_1\rVert < \delta'_{\varepsilon/2}, then \lvert S(f;\dot P_1) - L'\rvert < \varepsilon/2.Definition 7.1.1 applied to L' with tolerance \varepsilon/2 > 0. Also there exists \delta''_{\varepsilon/2} > 0 such that if \dot P_2 is any tagged partition with \lVert P_2\rVert < \delta''_{\varepsilon/2}, then \lvert S(f;\dot P_2) - L''\rvert < \varepsilon/2.Definition 7.1.1 applied to L'' with tolerance \varepsilon/2 > 0. Now let \delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0 and let \dot P be a tagged partition with \lVert P\rVert < \delta_\varepsilon.Minimum of two positive numbers is positive (M08); an equal partition with n > (b-a)/\delta_\varepsilon guarantees such a \dot P exists. Since both \lVert P\rVert < \delta'_{\varepsilon/2} and \lVert P\rVert < \delta''_{\varepsilon/2}, the Triangle Inequality gives
\lvert L' - L''\rvert = \lvert L' - S(f;\dot P) + S(f;\dot P) - L''\rvert \le \lvert L' - S(f;\dot P)\rvert + \lvert S(f;\dot P) - L''\rvert < \varepsilon/2 + \varepsilon/2 = \varepsilon.
Steps 2 and 3 both apply to the same \dot P; Triangle Inequality (P05). Since \varepsilon > 0 is arbitrary, it follows that L' = L''. ∎Arbitrary \varepsilon rule (P05): a non-negative number \lvert L' - L''\rvert < \varepsilon for all \varepsilon > 0 must be 0.   Where the hypothesis was used & Worked application (Ex 7.1.9) Where used: f \in \mathcal{R}[a,b] guarantees that one partition \dot P with \lVert P\rVert < \delta_\varepsilon satisfies the closeness bound for L' and L'' simultaneously. Worked example (Bartle Ex 7.1.9): If f \in \mathcal{R}[a,b] and (\dot P_n) is any sequence of tagged partitions with \lVert P_n\rVert \to 0, then \lim_{n\to\infty} S(f;\dot P_n) = \int_a^b f. Indeed, given \varepsilon > 0, take \delta_\varepsilon from Definition 7.1.1; since \lVert P_n\rVert \to 0, there is K \in \mathbb{N} with \lVert P_n\rVert < \delta_\varepsilon for all n \ge K, so \lvert S(f;\dot P_n) - \int_a^b f\rvert < \varepsilon for all n \ge K. By uniqueness of limits, \lim S(f;\dot P_n) must equal the unique integral \int_a^b f. In the model paper: Q4. </section>
<!-- T05 (K05) --> <section class="page"> T05<h2>Integrals straight from the definition: k, step g, and h(x)=x</h2> Three foundational calculations directly from Definition 7.1.1 (Bartle 7.1.4(a)–(c); PYQ MTS-25 Q4). <figure> {{fig:F15_step_jump_strip}} <figcaption>What to notice: For g(x)=2 on [0,1] and 3 on (1,3] with \lVert P\rVert < \delta, the subintervals with tag \le 1 cover [0,1-\delta] and lie inside [0,1+\delta], so the sum can only err in the \delta-strip around the jump x=1 by at most 5\delta. What it does not prove: The picture shows one \delta; the proof verifies \lvert S(g;\dot P)-8\rvert \le 5\delta < \varepsilon for every \varepsilon>0 via \delta_\varepsilon=\varepsilon/10.</figcaption> </figure> Examples 7.1.4(a), (b), (c) (Bartle pp. 202–204) (a) Constant function: Every constant function f(x) := k on [a,b] is in \mathcal{R}[a,b] and \int_a^b k = k(b-a). (b) Two-piece step function: g : [0,3] \to \mathbb{R} defined by g(x) := 2 for 0 \le x \le 1, and g(x) := 3 for 1 < x \le 3 is in \mathcal{R}[0,3] and \int_0^3 g = 8. (c) Identity function: h(x) := x on [0,1] is in \mathcal{R}[0,1] and \int_0^1 x\,dx = \tfrac12. In plain words: for a constant, every Riemann sum is exact; for a step function, the error comes only from the subinterval(s) straddling the jump; for x, midpoint tags telescope to 1/2 and any other tags differ

by less than \delta. Proof of 7.1.4(a) · Constant function (PYQ MTS-25 Q4) Let f(x) := k for all x \in [a,b]. For any tagged partition \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n of [a,b], S(f;\dot P) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = \sum_{i=1}^n k(x_i - x_{i-1}) = k\sum_{i=1}^n (x_i - x_{i-1}) = k(b - a).
Factor out k (P02) and telescope \sum (x_i - x_{i-1}) = x_n - x_0 = b - a (M01). Hence for any \varepsilon > 0, we can choose \delta_\varepsilon := 1 so that if \lVert P\rVert < \delta_\varepsilon, then \lvert S(f;\dot P) - k(b-a)\rvert = 0 < \varepsilon. Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] and \int_a^b k = k(b-a). ∎Definition 7.1.1 satisfied with L = k(b-a).   Proof of 7.1.4(b) · Step function g(x)=2 on [0,1], 3 on (1,3] Let \dot P be a tagged partition of [0,3] with \lVert P\rVert < \delta. Split \dot P into \dot P_1 (tags t_i \in [0,1], where g=2) and \dot P_2 (tags t_i \in (1,3], where g=3), so S(g;\dot P) = S(g;\dot P_1) + S(g;\dot P_2).Tag-splitting tool (M10). Let U_1 be the union of all subintervals in \dot P_1. Since \lVert P\rVert < \delta, we have [0, 1-\delta] \subseteq U_1 \subseteq [0, 1+\delta].If u \in [0,1-\delta] lies in I_k, then x_k \le u + \delta \le 1, so t_k \le 1 and I_k \in \dot P_1; conversely if t_k \le 1, then x_k \le t_k + \delta \le 1 + \delta (M10). Since g(t_i) = 2 on \dot P_1, the sum S(g;\dot P_1) equals 2 times the total length of U_1, so 2(1-\delta) \le S(g;\dot P_1) \le 2(1+\delta).Length of [0,1-\delta] is 1-\delta and length of [0,1+\delta] is 1+\delta. Similarly, the union U_2 of subintervals in \dot P_2 satisfies [1+\delta, 3] \subseteq U_2 \subseteq [1-\delta, 3] (lengths 2-\delta and 2+\delta), so 3(2-\delta) \le S(g;\dot P_2) \le 3(2+\delta).Same tag-inclusion argument on (1,3] with g(t_i) = 3. Adding the inequalities in Steps 3 and 4 gives 8 - 5\delta \le S(g;\dot P) \le 8 + 5\delta, or \lvert S(g;\dot P) - 8\rvert \le 5\delta.2(1-\delta)+3(2-\delta)=8-5\delta and 2(1+\delta)+3(2+\delta)=8+5\delta. Given \varepsilon > 0, choose \delta_\varepsilon := \varepsilon/10 > 0 (any \delta_\varepsilon < \varepsilon/5 works). If \lVert P\rVert < \delta_\varepsilon, then \lvert S(g;\dot P) - 8\rvert \le 5(\varepsilon/10) = \varepsilon/2 < \varepsilon. Thus g \in \mathcal{R}[0,3] and \int_0^3 g = 8. ∎Choosing \delta_\varepsilon from 5\delta < \varepsilon (M09).   </section>
<section class="page"> Scratch · Example 7.1.4(c): how to integrate h(x)=x on [0,1] from Definition 7.1.1 On subinterval I_i = [x_{i-1},x_i], choose the midpoint tag q_i = \tfrac12(x_{i-1}+x_i). Then h(q_i)(x_i - x_{i-1}) = \tfrac12(x_i + x_{i-1})(x_i - x_{i-1}) = \tfrac12(x_i^2 - x_{i-1}^2), which telescopes! What if the opponent picks arbitrary tags t_i \in I_i? Both t_i and q_i lie in I_i, so \lvert t_i - q_i\rvert \le x_i - x_{i-1} \le \lVert P\rVert < \delta. Proof of 7.1.4(c) · \int_0^1 x\,dx = 1/2 (Bartle pp. 203–204) Let \{I_i\}_{i=1}^n be any partition of [0,1], choose the midpoint q_i := \tfrac12(x_{i-1}+x_i) of each I_i = [x_{i-1},x_i], and let \dot Q := \{(I_i,q_i)\}_{i=1}^n. Then h(q_i)(x_i - x_{i-1}) = \tfrac12(x_i + x_{i-1})(x_i - x_{i-1}) = \tfrac12(x_i^2 - x_{i-1}^2).
Difference of two squares (x_i + x_{i-1})(x_i - x_{i-1}) = x_i^2 - x_{i-1}^2. Summing over i=1,\dots,n telescopes: S(h;\dot Q) = \sum_{i=1}^n \tfrac12(x_i^2 - x_{i-1}^2) = \tfrac12(x_n^2 - x_0^2) = \tfrac12(1^2 - 0^2) = \tfrac12.
Telescoping sum (M01) with x_0 = 0 and x_n = 1. <div

class="step">Now let \dot P = \{(I_i,t_i)\}_{i=1}^n be an arbitrary tagged partition of [0,1] with \lVert P\rVert < \delta, and let \dot Q have the same partition points with midpoint tags q_i. Since t_i, q_i \in I_i, we have \lvert t_i - q_i\rvert < \delta.Two points in the same subinterval of length \le \lVert P\rVert < \delta are within \delta of each other. By the Triangle Inequality,
\lvert S(h;\dot P) - S(h;\dot Q)\rvert \le \sum_{i=1}^n \lvert t_i - q_i\rvert(x_i - x_{i-1}) < \delta\sum_{i=1}^n (x_i - x_{i-1}) = \delta(x_n - x_0) = \delta.
Sum bound (M04) and telescoping \sum(x_i - x_{i-1}) = 1 - 0 = 1 (M01). Since S(h;\dot Q) = \tfrac12, we have \lvert S(h;\dot P) - \tfrac12\rvert < \delta whenever \lVert P\rVert < \delta. Given \varepsilon > 0, take \delta_\varepsilon := \varepsilon; then \lVert P\rVert < \delta_\varepsilon \Rightarrow \lvert S(h;\dot P) - \tfrac12\rvert < \varepsilon. Thus h \in \mathcal{R}[0,1] and \int_0^1 x\,dx = \tfrac12. ∎Definition 7.1.1 with \delta_\varepsilon = \varepsilon.   Worked extension · Q(x) = x^2 on [a,b], 0 \le a < b (Bartle Exercise 7.1.14) Choose special tags q_i := \sqrt{\tfrac13(x_i^2 + x_i x_{i-1} + x_{i-1}^2)}. Since x_{i-1}^2 \le \tfrac13(x_i^2 + x_i x_{i-1} + x_{i-1}^2) \le x_i^2, we have q_i \in [x_{i-1},x_i]. Then Q(q_i)(x_i - x_{i-1}) = \tfrac13(x_i^2 + x_i x_{i-1} + x_{i-1}^2)(x_i - x_{i-1}) = \tfrac13(x_i^3 - x_{i-1}^3), so S(Q;\dot Q) = \tfrac13(b^3 - a^3). For any tags t_i \in I_i, \lvert t_i^2 - q_i^2\rvert = \lvert t_i + q_i\rvert\lvert t_i - q_i\rvert < 2b\delta, so \lvert S(Q;\dot P) - \tfrac13(b^3-a^3)\rvert < 2b(b-a)\delta. Taking \delta_\varepsilon = \varepsilon/[2b(b-a)] proves \int_a^b x^2\,dx = \tfrac13(b^3 - a^3). In the model paper: Q5 (PYQ MTS-25 Q4). </section>
<!-- T06 (K06) --> <section class="page"> T06<h2>Changing a function at finitely many points</h2> Altering g \in \mathcal{R}[a,b] at finitely many points affects neither integrability nor the integral (Bartle 7.1.3, 7.1.4(d); PYQ MAT-22 MCQ 18, MAT-21 MCQ 13, MTS-22 MCQ 7). Theorem 7.1.3 (Bartle p. 201) · say it word for wordIf g is Riemann integrable on [a,b] and if f(x) = g(x) except for a finite number of points in [a,b], then f is Riemann integrable and \int_a^b f = \int_a^b g. In plain words: a single point (or finitely many points) has width zero, so changing the height of f at finitely many points changes any Riemann sum by at most a multiple of \lVert P\rVert, which vanishes as \lVert P\rVert \to 0. Hypotheses · Theorem 7.1.3 g \in \mathcal{R}[a,b]if g is Dirichlet C7 and f changes g at 5 points, f is still not integrable the exceptional set where f(x) \ne g(x) is finiteon a countably infinite set, integrability can fail: g(x)=0 is integrable on [0,1], but changing g on \{1/k : k\in\mathbb{N}\} to H(1/k)=k gives an unbounded function H \notin \mathcal{R}[0,1] (or changing g=0 on \mathbb{Q}\cap[0,1] to 1 gives Dirichlet C7) Skeleton · apply Theorem 7.1.3Observe that f(x) = g(x) for all x \in [a,b] \setminus \{c_1,\dots,c_m\}, a finite set, and g \in \mathcal{R}[a,b] with \int_a^b g = $?. By Theorem 7.1.3, $f \in R[a,b]$ and $\int_a^b f = \int_a^b g = $?. ∎ Proof of Theorem 7.1.3 (Bartle pp. 201–202) It suffices to prove the case of one exceptional point $c \in [a,b]$; the case of m points follows by mathematical induction, changing one point at a time.Standard induction on the number of

exceptional points. Let c be the exceptional point, let $L := \int_a^b g$, and assume $f(x) = g(x)$ for all $x \neq c$.Unpack the hypothesis. For any tagged partition $\dot{P}$, the sums $S(f;\dot{P})$ and $S(g;\dot{P})$ have identical terms except at most two (when c is a partition point tag of two adjacent subintervals). Hence $$\lvert S(f;\dot P) - S(g;\dot P)\rvert = \left\lvert\sum (f(t_i) - g(t_i))(x_i - x_{i-1})\right\rvert \le 2\big(\lvert g(c)\rvert + \lvert f(c)\rvert\big)\lVert P\rVert.$$At most 2 subintervals have tag c (M03), each of width $\le \lVert P\rVert$, and $\lvert f(c) - g(c)\rvert \le \lvert f(c)\rvert + \lvert g(c)\rvert$ (M04). (Note: Bartle prints $f(x_i) - g(x_i)$ inside the sum; the summand is evaluated at the tags t_i.) Given $\varepsilon > 0$, choose $\delta_1 > 0$ with $\delta_1 < \varepsilon/\big(4(\lvert f(c)\rvert + \lvert g(c)\rvert + 1)\big)$ (so $2(\lvert g(c)\rvert + \lvert f(c)\rvert)\delta_1 < \varepsilon/2$), and choose $\delta_2 > 0$ such that $\lVert P\rVert < \delta_2 \Rightarrow \lvert S(g;\dot P) - L\rvert < \varepsilon/2$.Definition 7.1.1 for g with tolerance $\varepsilon/2$, and M09 for the exceptional-point error. Let $\delta := \min\{\delta_1, \delta_2\} > 0$. If $\lVert P\rVert < \delta$, then $$\lvert S(f;\dot P) - L\rvert \le \lvert S(f;\dot P) - S(g;\dot P)\rvert + \lvert S(g;\dot P) - L\rvert < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$ Therefore $f \in R[a,b]$ with $\int_a^b f = L = \int_a^b g$. ∎$\varepsilon/2$ argument and Triangle Inequality (M08, P05).   Worked examples · Bartle Ex 7.1.6 & Example 7.1.4(d) 1. Bartle Ex 7.1.6: Let f(x) = 2 on [0,1) and 1 on [1,2]; let h(x) = 2 on [0,1), h(1) = 3, and h(x) = 1 on (1,2]. Both differ from the step function g(x)=2 on [0,1], 1 on (1,2] at most at the single point c = 1, so by Theorem 7.1.3, \int_0^2 h = \int_0^2 f = 2(1) + 1(1) = 3. 2. Countably many points where heights shrink to 0 (Bartle Example 7.1.4(d)): Let G(1/n) = 1/n for n \in \mathbb{N} and G(x) = 0 elsewhere on [0,1]. Theorem 7.1.3 does NOT apply directly (infinitely many points!), but given \varepsilon > 0, the set E = \{x : G(x) \ge \varepsilon\} = \{1, \tfrac12, \dots, \tfrac1n\} (where n \le 1/\varepsilon) is finite with n points. Taking \delta = \varepsilon/(2n), any \dot P with \lVert P\rVert < \delta splits into \dot P_0 (tags outside E, where G < \varepsilon and total width \le 1) and \dot P_1 (tags in E, at most 2n intervals of width <\delta with G \le 1), giving 0 \le S(G;\dot P) < \varepsilon(1) + (2n)\delta(1) = 2\varepsilon. Hence G \in \mathcal{R}[0,1] and \int_0^1 G = 0. Exam trap · PYQ MTS-22 MCQ 7 vs MAT-22 MCQ 18 If f(x) = 0 except at finitely many points c_1,\dots,c_n \in [a,b], then by Theorem 7.1.3 (with g \equiv 0) both f \in \mathcal{R}[a,b] and \int_a^b f = 0 hold (MAT-22 MCQ 18). If f(x) = g(x) off the infinite set \{1/n : n\in\mathbb{N}\} (MTS-22 MCQ 7), knowing f \in \mathcal{R}[0,1] does NOT force g \in \mathcal{R}[0,1] unless g is also bounded (take f=0 and g(1/n)=n). In the model paper: Q6 (PYQ MAT-22 MCQ 18, MAT-21 MCQ 13, MTS-22 MCQ 7). </section>
<!-- T07 (K07) --> <section class="page"> T07<h2>Linearity and monotonicity of the integral</h2> Integrals respect scalar multiples, sums, and \le inequalities (Bartle Theorem 7.1.5; PYQ MTS-25 Q5, MAT-21 Q29, MTS-22 Q5, MAT-22 MCQ 14). Theorem 7.1.5 · Properties of the Integral (Bartle p. 204)Suppose f and g are in \mathcal{R}[a,b]. Then: (a) If k \in \mathbb{R}, the

function kf is in \mathcal{R}[a,b] and \int_a^b kf = k\int_a^b f. (b) The function f + g is in \mathcal{R}[a,b] and \int_a^b (f+g) = \int_a^b f + \int_a^b g. (c) If f(x) \le g(x) for all x \in [a,b], then \int_a^b f \le \int_a^b g. In plain words: because every Riemann sum is a finite sum that is linear and order-preserving, passing to the integral preserves constants, sums, and \le. Hypotheses · Theorem 7.1.5 both f \in \mathcal{R}[a,b] and g \in \mathcal{R}[a,b]f = \text{Dirichlet} C7 and g = 1 - f have f+g = 1 \in \mathcal{R}[0,1], yet neither \int_0^1 f nor \int_0^1 g exists in (c), f(x) \le g(x) on [a,b] (or all but finitely many points by 7.1.3)if f \le g fails on an interval of positive length where f > g, \int_a^b f \le \int_a^b g fails Proof of Theorem 7.1.5 (Bartle pp. 204–205) For any tagged partition \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n of [a,b], finite sum rules (P02) give
$$S(kf;\dot P) = kS(f;\dot P), \qquad S(f+g;\dot P) = S(f;\dot P) + S(g;\dot P),$$
and since x_i - x_{i-1} > 0, f(x) \le g(x) on [a,b] implies S(f;\dot P) \le S(g;\dot P).Linearity and monotonicity of finite sums (P02). Part (a): If k = 0, kf = 0 \in \mathcal{R}[a,b] with integral 0 = 0\int_a^b f. If k \ne 0, given \varepsilon > 0 let \delta_\varepsilon > 0 be such that $\lVert P\rVert < \delta_\varepsilon \Rightarrow \lvert S(f;\dot P) - \int_a^b f\rvert < \varepsilon/\lvert k\rvert$; then $\lvert S(kf;\dot P) - k\int_a^b f\rvert = \lvert k\rvert\,\lvert S(f;\dot P) - \int_a^b f\rvert < \varepsilon$.Definition 7.1.1 applied with tolerance \varepsilon/\lvert k\rvert. Part (b): Given \varepsilon > 0, choose \delta_1, \delta_2 > 0 for f, g with tolerance \varepsilon/2, and put \delta_\varepsilon := \min\{\delta_1, \delta_2\} > 0. If $\lVert P\rVert < \delta_\varepsilon$, then both $\lvert S(f;\dot P) - \int_a^b f\rvert < \varepsilon/2$ and $\lvert S(g;\dot P) - \int_a^b g\rvert < \varepsilon/2$ hold.Common \delta_\varepsilon argument (M08) as in Theorem 7.1.2. By Step 1 and the Triangle Inequality,
$$\left\lvert S(f+g;\dot P) - \left(\int_a^b f + \int_a^b g\right)\right\rvert \le \left\lvert S(f;\dot P) - \int_a^b f\right\rvert + \left\lvert S(g;\dot P) - \int_a^b g\right\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$
Since \varepsilon > 0 is arbitrary, f + g \in \mathcal{R}[a,b] with integral \int_a^b f + \int_a^b g.Triangle Inequality (P05) and Definition 7.1.1. Part (c): For $\lVert P\rVert < \delta_\varepsilon$ as in Step 3, unpacking the absolute values gives
$$\int_a^b f - \frac{\varepsilon}{2} < S(f;\dot P) \le S(g;\dot P) < \int_a^b g + \frac{\varepsilon}{2}.$$
Adding \varepsilon/2 yields \int_a^b f < \int_a^b g + \varepsilon. Since \varepsilon > 0 is arbitrary, \int_a^b f \le \int_a^b g. ∎Unpacking $\lvert S - L\rvert < \varepsilon/2$ and the arbitrary \varepsilon rule (P05).   Worked example · Bounding \lvert\int_a^b f\rvert (Bartle Exercise 7.1.8) & MCQ trap Bartle Ex 7.1.8: If f \in \mathcal{R}[a,b] and \lvert f(x)\rvert \le M for all x \in [a,b], then -M \le f(x) \le M on [a,b]. By Theorem 7.1.5(c) and Example 7.1.4(a), -M(b-a) = \int_a^b (-M) \le \int_a^b f \le \int_a^b M = M(b-a), which means \left\lvert\int_a^b f\right\rvert \le M(b-a). MCQ trap (MAT-22 MCQ 14): f(x) \le g(x) implies \int_a^b f \le \int_a^b g (with \le), never strict < in general (for instance, take f = g, or take f=0 and g(0)=1, g(x)=0 for x\in(0,1], where f \le g and f(0)<g(0) yet \int_0^1 f = 0 = \int_0^1 g). In the model paper: Q21 (PYQ MAT-21 Q29, MTS-22 Q5, MTS-25 Q5, MAT-22 MCQ 14). </section>

<!-- T08 (K08) --> <section class="page"> T08<h2>The Boundedness Theorem</h2> Every Riemann integrable function is bounded; unbounded \Rightarrow \notin \mathcal{R}[a,b] (Bartle Theorem 7.1.6; PYQ MAT-22 MCQ 15, MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9). Theorem 7.1.6 · Boundedness Theorem (Bartle p. 205) · say it word for wordIf f \in \mathcal{R}[a,b], then f is bounded on [a,b]. In plain words: if f blows up to \pm\infty anywhere on [a,b], you can pick a tag on that subinterval to make a single rectangle arbitrarily tall, destroying \lvert S(f;\dot P) - L\rvert < 1. Direction & Converse · Theorem 7.1.6 Necessary condition: f \in \mathcal{R}[a,b] \Rightarrow f is bounded on [a,b] (contrapositive: f unbounded on [a,b] \Rightarrow f \notin \mathcal{R}[a,b])Converse is FALSE: bounded does NOT imply integrable! Dirichlet's function C7 satisfies 0 \le f(x) \le 1 on [0,1] (bounded), yet f \notin \mathcal{R}[0,1]. Also f \in \mathcal{R}[a,b] does NOT imply f is continuous or monotone (C8 Thomae). Knowf \in \mathcal{R}[a,b] with integral L. Assume for contradiction that f is unbounded on [a,b].WantConstruct a tagged partition \dot Q with \lVert Q\rVert < \delta but \lvert S(f;\dot Q)\rvert > \lvert L\rvert + 1, contradicting Definition 7.1.1 for \varepsilon = 1. Proof of Theorem 7.1.6 (Bartle pp. 205–206) Assume that f is an unbounded function in \mathcal{R}[a,b] with integral L. For \varepsilon = 1, there exists \delta > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta, then \lvert S(f;\dot P) - L\rvert < 1, which by the Triangle Inequality implies \lvert S(f;\dot P)\rvert \le \lvert S(f;\dot P) - L\rvert + \lvert L\rvert < \lvert L\rvert + 1.
Definition 7.1.1 with \varepsilon = 1 and Triangle Inequality (P05). Let Q = \{[x_{i-1},x_i]\}_{i=1}^n be a fixed partition of [a,b] with \lVert Q\rVert < \delta. Since f is unbounded on [a,b], f must be unbounded on at least one subinterval [x_{k-1},x_k].If f were bounded on each [x_{i-1},x_i] by M_i, it would be bounded on all of [a,b] by \max\{M_1,\dots,M_n\}. We now tag Q as follows: for i \ne k, fix t_i := x_i. Since f is unbounded on [x_{k-1},x_k] and x_k - x_{k-1} > 0, there exists a tag t_k \in [x_{k-1},x_k] such that \lvert f(t_k)(x_k - x_{k-1})\rvert > \lvert L\rvert + 1 + \left\lvert\sum_{i\ne k} f(t_i)(x_i - x_{i-1})\right\rvert.
For i \ne k the sum is now a fixed number; \lvert f(t_k)\rvert can be chosen arbitrarily large on [x_{k-1},x_k]. By the Reverse Triangle Inequality \lvert A + B\rvert \ge \lvert A\rvert - \lvert B\rvert, we obtain
\lvert S(f;\dot Q)\rvert \ge \lvert f(t_k)(x_k - x_{k-1})\rvert - \left\lvert\sum_{i\ne k} f(t_i)(x_i - x_{i-1})\right\rvert > \lvert L\rvert + 1,
which contradicts Step 1. Therefore f must be bounded on [a,b]. ∎Reverse Triangle Inequality (P05) yields the contradiction.   Worked examples · H(1/k)=k (PYQ MTS-22 MCQ 9, MAT-21 MCQ 17) & Bartle Ex 7.1.10 1. H(1/k) = k (k \in \mathbb{N}), H(x) = 0 elsewhere on [0,1]: Since H(1/k) = k \to \infty as k \to \infty, H is unbounded on [0,1]. By Theorem 7.1.6, H \notin \mathcal{R}[0,1] immediately! (Notice the contrast with Example 7.1.4(d) where G(1/n) = 1/n stays bounded and has integral 0.) 2. Why one convergent sequence of Riemann sums does not prove integrability (Bartle Ex 7.1.10): Let g(x) = 0 for rational x \in [0,1] and g(x) = 1/x for irrational x \in [0,1]. Since g is unbounded near 0, g \notin \mathcal{R}[0,1] by

Theorem 7.1.6. Yet if we pick the equal partition P_n with rational tags $t_i = i/n$, then $S(g;\dot P_n) = 0 \to 0$! You must control all tags, not just your favourite ones. In the model paper: Q7 (PYQ MAT-22 MCQ 15, MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9). </section>
<!-- T09 (K09) --> <section class="page"> T09<h2>Thomae's function is integrable (with integral 0)</h2> Discontinuous at every rational, continuous at every irrational, not monotone—yet in $\mathcal{R}[0,1]$ (Bartle Example 7.1.7). <figure>
{{fig:F09_thomae_popcorn}} <figcaption>What to notice: For Thomae's function $h(m/n)=1/n$, only 11 points in $[0,1]$ reach or exceed height $\varepsilon/2 = 1/5$ (circled set E); every other point sits inside the shaded band $h(x) < 1/5$. What it does not prove: Drawing up to denominator $n=5$ shows the pattern, whereas the proof works for arbitrary $\varepsilon>0$ because $1/n \ge \varepsilon/2 \iff n \le 2/\varepsilon$ always leaves finitely many fractions.</figcaption> </figure> Example 7.1.7 · Thomae's Function C8 (Bartle p. 206)Define $h : [0,1] \to \mathbb{R}$ by $h(x) := 0$ if $x \in [0,1]$ is irrational, $h(0) := 1$, and $h(x) := 1/n$ if $x \in [0,1]$ is the rational number $x = m/n$ where $m, n \in \mathbb{N}$ have no common integer factor except 1. Then h is continuous at every irrational point and discontinuous at every rational point of $[0,1]$, is not monotone on any subinterval, and $h \in \mathcal{R}[0,1]$ with $\int_0^1 h = 0$. In plain words: for any threshold $\varepsilon/2 > 0$, only finitely many rationals m/n have denominator $n \le 2/\varepsilon$. We trap those finitely many tall spikes in tiny subintervals of total width $\le \varepsilon/2$, and everywhere else the height is $<\varepsilon/2$. Know$0 \le h(x) \le 1$ on $[0,1]$; for any $\varepsilon > 0$, $E = \{x \in [0,1] : h(x) \ge \varepsilon/2\}$ is a finite set of n points.WantFind $\delta > 0$ so that $\lVert P\rVert < \delta \Rightarrow 0 \le S(h;\dot P) < \varepsilon$. Scratch · how we choose δSplit $\dot P = \dot P_1 \cup \dot P_2$ where $\dot P_1$ has tags in E (n points $\Rightarrow$ at most $2n$ subintervals). On $\dot P_1$: height ≤ 1, total width $\le 2n\delta$; want $2n\delta \le \varepsilon/2 \Leftarrow \delta = \varepsilon/(4n)$. On $\dot P_2$: height $< \varepsilon/2$, total width ≤ 1, so sum $< \varepsilon/2$. Total $< \varepsilon$. Proof of Example 7.1.7 (Bartle p. 206) Let $\varepsilon > 0$ be given. Since $h(m/n) = 1/n \ge \varepsilon/2 \iff n \le 2/\varepsilon$, there are only finitely many denominators n and finitely many numerators $m \le n$, so the set $E := \{x \in [0,1] : h(x) \ge \varepsilon/2\}$ is finite.Archimedean property (P13): only finitely many $n \in \mathbb{N}$ satisfy $n \le 2/\varepsilon$. Let n be the number of points in E and set $\delta := \varepsilon/(4n) > 0$. Let $\dot P$ be any tagged partition of $[0,1]$ with $\lVert P\rVert < \delta$, and split $\dot P$ into $\dot P_1$ (tags $t_i \in E$) and $\dot P_2$ (tags $t_i \notin E$).Tag-splitting tool (M10). Allowing for a point of E being an endpoint tag of two adjacent subintervals, $\dot P_1$ has at most $2n$ subintervals, of total length at most $2n\delta = 2n\cdot\frac{\varepsilon}{4n} = \frac{\varepsilon}{2}$. Since $0 < h(t_i) \le 1$ on $\dot P_1$, we have $S(h;\dot P_1) \le 1 \cdot 2n\delta = \varepsilon/2$.Endpoint tag count (M03) and sum bound (M04). For tags t_i in $\dot P_2$, we have $h(t_i) < \varepsilon/2$, and the total length of the subintervals in $\dot P_2$ is at most 1 (the length of $[0,1]$), so $S(h;\dot P_2) < (\varepsilon/2)\cdot 1 = \varepsilon/2$.Definition of E: outside E, $h(x) < \varepsilon/2$. Combining Steps 3 and 4 gives $0 \le S(h;\dot P) = S(h;\dot P_1) + S(h;\dot P_2) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} =$

\varepsilon.
Since \varepsilon > 0 is arbitrary, h \in \mathcal{R}[0,1] and \int_0^1 h = 0. ∎Definition 7.1.1 with L = 0.   Check yourself · Bartle p. 206 numerical check List all points in E = \{x \in [0,1] : h(x) \ge 1/5\} (when \varepsilon/2 = 1/5, i.e. \varepsilon = 2/5), count them, and give Bartle's \delta. We need denominators 1,2,3,4,5 in lowest terms (plus 0, where h(0)=1): E = \{0, 1, \tfrac12, \tfrac13, \tfrac23, \tfrac14, \tfrac34, \tfrac15, \tfrac25, \tfrac35, \tfrac45\}, which has n = 11 points. Thus \delta = \varepsilon/(4n) = (2/5)/44 = 1/110. In the model paper: Q22. </section>
=== NOTES / DOUBTS (STAGE 1 OF 6) ===

- Missing attachments for Stage 6 (figs_m3.py): Only module3-references.txt was attached to this turn; figs.py and figs_m2.py were not attached. Please paste or attach figs.py and figs_m2.py before or at Stage 6 so src/m3/figs_m3.py can use the exact S and Plot class methods and helpers.
- Stage plan (6 stages total): Stage 1 (above): src/m3/prepare_01_start.html + src/m3/prepare_02_sec71.html (Cover, How-to, P01–P14, M01–M10, T01–T09); Stage 2: src/m3/prepare_03_sec72.html (T10–T19, §7.2); Stage 3: src/m3/prepare_04_sec73.html (T20–T32, §7.3 + Counterexample library + Which tool? drill + Theorem sheet + Recall plan + PYQ table); Stage 4: src/m3/paper.html (40-question model paper + coverage table); Stage 5: src/m3/answers_1.html + src/m3/answers_2.html (Q1–Q28 solutions); Stage 6: src/m3/answers_3.html (Q29–Q40 solutions) + src/m3/figs_m3.py.
- Topic IDs: Used two-digit zero-padded T01–T32, P01–P14, M01–M10 matching the Module 2 cover (src/m2/prepare_01_start.html) and content_m3.md.
- Bartle 7.1.3 proof typo note: Bartle's printed sum writes f(x_i)-g(x_i); our proof writes f(t_i)-g(t_i) and notes the book's print.